



Wave 'n' Wire: The Smart System Challenge

~25 TEAMS • 3–4 MEMBERS PER TEAM • 29 OCTOBER

Welcome to Wave 'n' Wire

25 → **10** → **5**
TEAMS BEGIN PHASE 1 REACH THE FINALE TAKE THE PODIUM

PHASE 1

Analyse & Architect

PAPER DESIGN STAGE // SCORED, NON-ELIMINATING

- Analyse the assigned scenario and identify required components
- Plan the overall system blueprint
- Design the MQTT topic hierarchy for the system
- Define the publisher / subscriber / payload structure

TYPE	POINTS
Component Identification	5
Planning / System Blueprint	5
MQTT Topic Architecture	5
Publisher / Subscriber / Payload Design	5
Total	20

All teams advance to Phase 2 — this stage is scored but not eliminating.

ALL TEAMS ADVANCE TO PHASE 2



PHASE 2

Build & Integrate

4 PROGRESSIVE LEVELS // SAME 6-STAGE WORKFLOW, INCREASING DIFFICULTY

1. Component Recognition

- Train a live computer-vision model using Google Teachable Machine and their own webcam to recognize the physical component cards required for that level

2. Circuit Builder

- Assemble the system on the platform's drag-and-drop virtual breadboard, with real-time validation of GPIO, power and ground connections

3. Command Configuration

- Map recognized components into device commands, wiring the circuit into the MQTT architecture designed in Phase 1

4. Gesture Model

- Train and attach a second Teachable Machine model for gesture recognition, so hand gestures can trigger real device actions in the simulation

5. Test Run

- A practice attempt against simulated inputs before the scored run

6. Final Simulation

- The scored, one-shot attempt: the full recognition → circuit → MQTT → AWS IoT Core → dashboard pipeline runs live, updating a real-time smart-environment dashboard

- Each level worth 20 marks, split across mandatory core tasks, scored secondary tasks, and an efficiency/constraint bonus
- Advancing requires completing all mandatory tasks and meeting the minimum score threshold per level (default 60%, admin-configurable)

Finalizing a level's Final Simulation locks it permanently — no resubmission. One emergency-component request is allowed per team, at a -5 mark penalty. Time does not add or subtract marks — used only as a tie-breaker. Completing Level 4 ends that team's run.

LEVELS 1–4 FOLLOW THE SAME SUBMISSION RULES



PROGRESSION

Level-Wise Submissions & Closing

SCORE-BASED RANKING // TIME USED ONLY AS A TIE-BREAKER

- Each level has its own constraints to be met and tasks that must be completed
- A team can progress to the next level only after the current level has been submitted
- Once Level 4 is submitted, the challenge is completed
- Teams are evaluated by the points they've scored
- Top 10 teams will reach the Finale

TOP 10 TEAMS ADVANCE TO THE FINALE



FINALE

The Finale

TOP 10 COMPETE FOR THE PODIUM

- Winner & Runner-Up are crowned from the two highest-scoring teams among the 10 finalists
- The remaining eight finalists compete for the final three podium spots — 3rd, 4th and 5th place

TOP 5 TEAMS TAKE THE PODIUM

EVENT GUIDE

WAVE 'N' WIR // DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING // 29 OCTOBER

PHASE	TITLE	FORMAT & MECHANICS	OUTCOME
Phase 1	Analyse & Architect	Scenario analysis, component identification, system blueprint, MQTT topic & pub/sub/payload design	Scored out of 20, non-eliminating
Phase 2	Build & Integrate	4 progressive levels, each the same 6-stage workflow (Component Recognition → Circuit Builder → Command Configuration → Gesture Model → Test Run → Final Simulation) at increasing difficulty	Scored out of 80
Finale	The Finale	Top 10 teams by score compete for the podium — two highest scorers become Winner and Runner-Up, remaining eight contest 3rd–5th place	Top 5 podium finishers

Wave 'n' Wire: The Smart System Challenge

Wave 'n' Wire: The Smart System Challenge puts ~25 teams through a phased build of a connected smart-environment system — from architecting an MQTT-based communication design on paper in Phase 1, to a six-stage build pipeline of computer-vision recognition, virtual circuitry, gesture control and live AWS IoT Core simulation across four levels in Phase 2. Every team advances on its own merit, and the event closes once organizers judge the right moment has come.

QUICK OVERVIEW — THE FLOW

~25 Teams
3–4 MEMBERS EACH

3 Stages
PHASE 1 / PHASE 2 / FINALE

Top 5
TAKE THE PODIUM

EVENT ROUNDS

Phase 1 — Analyse & Architect

- Identify required components for the assigned scenario
- Plan the system blueprint
- Design the MQTT topic hierarchy and publisher/subscriber/payload structure

ALL TEAMS ADVANCE — NON-ELIMINATING

Phase 2 — Build & Integrate

- 4 levels, each a Component Recognition → Circuit Builder → Command Configuration → Gesture Model → Test Run → Final Simulation pipeline
- Mandatory core tasks, scored secondary tasks, and an efficiency/constraint bonus per level
- One emergency-component request allowed per team, at a -5 mark penalty

TOP 10 TEAMS ADVANCE TO THE FINALE

Finale — The Podium Round

- Top 10 teams by score compete for the podium
- Two highest-scoring teams are crowned Winner and Runner-Up
- Remaining eight finalists contest 3rd, 4th and 5th place

TOP 5 TEAMS TAKE THE PODIUM

COORDINATORS

STUDENT COORD.

Hemanth C
Dept. of ECE

B U Ravi Thejas
Dept. of ECE

Kaushal Rao A
Dept. of ECE

FACULTY COORD.

Mrs. Swetha K
Assistant Professor, ECE

Mrs. Swetha B V
Assistant Professor, ECE

ORGANIZING TEAM

Ankitha Jade
AWS SBG Core Member
+91 6360024049

Sadhana S
AWS SBG Core Member
+91 91418 00861